

TimeTensors Experiment 4: Normalization Layers

1 Theoretical overview and goal

Normalization can improve conditioning and remove distribution shifts. Global standardization uses training-wide (μ, σ) ; global min-max uses training extrema. Instance methods instead estimate statistics from each look-back and invert the transform after forecasting. RevIN adds a learnable affine map to this reversible instance transform.

Experimental goal. Compare global and per-window alignment, distinguish standardization from range normalization, and test whether RevIN’s affine parameters add value for DLinear and PatchTST.

2 Implementation details

The methods are identity, global standard, global min-max, instance min-max, non-affine instance normalization, and affine RevIN. Global statistics are computed only from accessible training look-backs to prevent leakage. Instance statistics are computed independently for each sample and reused to restore predictions to data space.

The default benchmark uses six datasets, three settings, seeds 1–3, and a DLinear table. Training defaults are random sampling, batch size 256, learning rate 10^{-5} , 10,000 optimizer steps, and 1,000-step validation. PatchTST is the confirmatory second stage. The launcher is `03_normalizations.slurm`.

Report test MSE after inverse normalization, with seed standard deviation and explicit row multiplier. Keep the configured training loss fixed across the normalization sweep. Save fitted global statistics with the run so the comparison is reproducible.

3 Interpretation

Global methods test optimization conditioning but cannot adapt to shifted test levels. Instance methods trade absolute level information for invariance. Differences between non-affine instance normalization and RevIN isolate the learned affine component, not the reversible transform itself.